Ecole Supérieure
d'Informatique et du Numérique
COLLEGE OF ENGINEERING & ARCHITECTURE

Lab 4: Applied Crypto & B.C
Instructor: Prof. Oumaima FADI
Level: Cybersecurity - 4th Year

Lab 3: Feistel Structure and the DES Algorithm

Objectives

- Understand the logic and reversibility of the Feistel cipher structure.
- Identify the main components and workflow of DES.
- Explain the concepts of **confusion** and **diffusion** in symmetric cryptography.
- Analyze the impact of key order, permutations, and parity bits on DES encryption/decryption.

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Instructions

- The lab report must be submitted one week after the lab session on the Moodle platform, respecting the deadline mentioned there.
- The lab must be performed individually in class, but the report should be submitted in groups of up to 2 students.
- Lab groups must remain the same for all reports throughout the semester.

Exercise 1 – The Feistel Principle

1. Explain the difference between encryption and decryption in a Feistel network. Why can both use the same function F?

2. Consider a 3-round Feistel cipher defined by:

$$L_i = R_{i-1}, R_i = L_{i-1} \oplus f(R_{i-1}, K_i)$$

   Show how $L_0$and R_0can be recovered from $L_3$and $R_3$by applying the round keys in reverse order.

3. What happens if the decryption process uses the same order of subkeys as encryption (i.e., not reversed)?

4. Explain why a Feistel cipher is reversible even if the round function $f$itself is not invertible.

Exercise 2 – The Structure of DES

1. What are the block size and the effective key size used in DES?
2. DES applies 16 Feistel rounds using subkeys $K_1, K_2, \ldots, K_{16}$. How does the decryption process use these keys?
3. Describe the steps of the DES round function f:
   - Expansion (E-box)
   - XOR with the subkey
   - Substitution (S-boxes)
   - Permutation (P-box)

     Explain briefly how these steps contribute to confusion and diffusion.

Exercise 3 – Parity Bits

1. Why is DES said to use a 64-bit key, even though only 56 bits are effectively used?
2. What is the purpose of the remaining 8 bits?
3. What could happen if one parity bit is changed during key transmission?

Exercise 4 – Simplified Feistel Example

Consider a 1-round Feistel cipher where:

$$L_0 = 1010,$$
$$R_0 = 1100,$$
$$K_1 = 1111$$
$$K_2 = 1001$$

and the round function is $f(R, K) = R \oplus K$.

1. Compute $L_1$and R_1(encryption) and $L_2$and R_2
2. Show how to recover $L_0, R_0$from L_2, R_2(decryption).
3. Repeat the process without reversing the key and observe what happens. What does this show ?

Homework – Conceptual Analysis

1. Explain in your own words the difference between confusion and diffusion.
2. Identify which parts of DES mainly provide confusion and which provide diffusion.
3. Why are both properties essential to resist statistical or differential attacks?